

DoangJoo "Alan" Synn

Ph.D. Student in Computer Science · Georgia Institute of Technology

Advised by [Prof. HyunJoo Oh](#) and [Prof. Sehoon Ha](#)

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Research Interests

My research explores computational design and creativity, with a focus on representing and supporting creative intent in generative and interactive systems. I build models and interfaces that translate high-level human intent into expressive, controllable outcomes, spanning domains such as computer graphics, motion synthesis, and kinematic design.

Education

[Georgia Tech](#), Ph.D. in Computer Science

2022–Present

- [School of Interactive Computing](#); advised by Profs. [Oh](#) & [Ha](#)
- Research: computational design, mechanical artifacts, and motion synthesis (HCI/Graphics)

[Korea University](#), M.Eng. in ECE

2020–2022

- Thesis: *On Efficient Data Delivery for Machine Learning Systems*

[Korea University](#), B.Eng. in EE

2012–2020

Experience

Ph.D. Research Intern, [Dolby Laboratories](#)

2024–2025

- Built a time-series, multimodal datastore supporting low-latency media analytics in the Experience Delivery Lab.

MLOps Engineering Intern, [LG Uplus Corp.](#)

2023

- Led [DeepCrunch](#), a model-compression library supporting multi-backend AI inference for efficient edge deployment.

Research Scientist, [Protopia AI Inc.](#)

2021–2022

- Built [Stained-Glass](#), a privacy-preserving deep-learning framework enabling inference without raw-data exposure.

Cloud Architect · [AWS Educate](#) Cloud Ambassador

2020–2022

- Led engineering for four startup services, translating ambiguous product requirements into production backend systems.

Server Engineer, [Flysher Inc.](#)

2019

- Shipped backend features for four social-game titles; one reached Facebook's global top 10.

System Engineer, [AKA Intelligence](#) ([Musio](#))

2017–2019

- Designed embedded Linux boards and firmware for [Musio](#), a deployed social robot.

Selected Publications

1 [D. Synn](#), Z. Guo, S. Ha, and H. Oh, “MotionSmith: A Sketch-Based Design System for Automata Making,” *ACM CHI*, Apr. 2026.

2 J. Yu, P. Garg, [D. Synn](#), and H. Oh, “Tangible-MakeCode: Bridging Physical Coding Blocks with a Web-Based Programming Interface for Collaborative and Extensible Learning,” *ACM CHI*, Apr. 2025.